

Gowtham kumar

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Professional Summary:

- 6+ Years of extensive experience in designing, developing, and optimizing data pipelines, ETL processes, and data warehouses.
- Extensive experience with different phases of the project lifecycle, including project initiation, requirement gathering, system design, coding, testing, and debugging client-server-based applications.
- Experience in Azure Cloud Computing services, such as Azure Data Factory, Azure Storage accounts, Azure Synapse, Key Vaults, ADLS Gen2, Azure App Insights, Azure Databricks, ARM templates, Azure DevOps, Integration Runtimes, Azure Data Flows.
- Proficient in applying SDLC software development processes and establishing a business analysis methodology.
- Lead data mapping, exploratory data analysis, and logical data modeling between source and destination systems.
- Designed Azure architecture, Cloud migration, and event processing using Azure Data Factory.
- Designed solutions that addressed complex data challenges, ensuring alignment with organizational goals and scalability requirements.
- Proficient in designing data pipelines for streaming web data and RDBMS source data.
- Merit of building ETL architecture and source-to-target mapping for data loading into data warehouses.
- Configured geo-redundant storage (GRS) and point-in-time restore (PITR) for Azure Blob Storage and Azure Data Lake Gen2 to protect against regional outages.
- Developed disaster recovery (DR) plans using Azure Site Recovery and failover groups for Azure SQL Managed Instances, minimizing downtime during outages.
- Automated log backups and snapshots for Azure VMs running SQL Server using Azure Automation and PowerShell scripts.
- Developed T-SQL stored procedures in Azure Synapse dedicated SQL pools to drive data transformation logic for ADF pipelines.
- Used ADF Lookup and Stored Procedure activities to parameterize and execute T-SQL scripts dynamically within pipelines.
- Applied T-SQL based business rules and validations as pre-load checks in ADF pipelines before loading data into curated layers.
- Worked with multiple join statements to retrieve data from multiple tables.
- Proficient in creating reports, dashboards, and visualizations using Power BI, implementing role-based security, schedule refresh, and performing data transformations using Power Query Editor.

Technical Skills

RDBMS: SQL Server 2019, 2017, 2016, 2014, 2012, Oracle

SDLC: Agile

Data Modeling: ER-win, MS Visio

Cloud Services: Azure

Development Tools: SSMS, SSAS (OLAP Cubes, Tabular Model)

Reporting Tools: SSRS, Power BI, Excel, Tableau

Source Control & Collaboration Tool: Team Foundation Server (TFS), SharePoint

Professional Experience

Persistent systems (client: Optum), Minneapolis, MN
Data engineer with Power BI

Nov 2023 to Present

- Developed data mapping, data governance, and cleansing rules for OLTP/OLAP and Master Data Management architectures.
- Built and maintained Azure Data Factory pipelines for moving data from flat files and SQL Server into Azure Data Lake using U-SQL and other native connectors.
- Used Azure Data Factory and Azure Synapse to integrate on-premises and cloud-based data sources in a virtualized environment.
- Utilized MongoDB for NoSQL database management, ensuring efficient storage and retrieval of unstructured data.
- Developed Python scripts to automate data extraction and processing tasks, enhancing the overall efficiency of the ETL pipelines.
- Worked on creating case management reports using Azure Data Lake Gen2 and developing ETL processes with SQL Server Integration Services (SSIS).
- Integrated Azure Event Hubs and Azure Stream Analytics for ingesting and processing real-time streaming data into Azure Data Lake and downstream analytics systems.
- Designed and maintained Azure Data Factory pipelines, using Copy Activity, Mapping Data Flows, and Pipeline Orchestrator to move and transform data from OLTP systems into curated layers.
- Designed and implemented complex T-SQL queries with multiple joins, CTEs, and window functions to support analytics and reporting.
- Created and maintained T-SQL functions (scalar and table-valued) to encapsulate business rules and reusable logic.
- Developed dynamic SQL scripts for parameterized execution of ETL processes and ad-hoc reporting needs.
- Applied query optimization techniques including indexing, statistics updates, and query rewrites to improve runtime efficiency.
- Built T-SQL scripts for data validation and reconciliation, ensuring data accuracy across staging, warehouse, and reporting layers.
- Automated batch data loading and updates with T-SQL jobs scheduled in SQL Server Agent.
- Created role-based security structures in SQL Server using schemas, roles, and GRANT/DENY statements.
- Used temporary tables and table variables to optimize intermediate result storage in complex ETL workflows.
- Used Great Expectations within Databricks to build data quality validation checks and enforce consistency across OLTP ingestion pipelines.
- Collaborated with business and analytics teams to clean, standardize, and reconcile OLTP datasets, ensuring readiness for downstream BI and reporting solutions.
- Orchestrated and automated Kubernetes deployments for scalable containerized applications and managed CI/CD pipelines for machine learning models.
- Implemented UPSERT pipeline for Change Data Capture using Audit fields like AUD_INSRT_TMSTP and AUD_UPDT_TMSTP for updating and inserting latest data from source to Target.

Loreal, NYC
Data engineer with Power BI

Jan 2023 to Sep 2023

- Designed and implemented logical and physical data models using Erwin and T-SQL, creating tables, views, and stored procedures to meet business requirements.
- Created data mapping, data governance, transformation, and cleansing rules involving OLTP and OLAP systems.

- Worked on building predictive models for inventory management and demand forecasting using AEP in combination with Azure Machine Learning.
- Identified dimensions, measures, and facts on OLTP sources, ensuring effective data integration processes for ETL and data presentation.
- Leveraged DBT to manage data transformations and enforce modular, reusable SQL development practices in cloud data warehouses.
- Used PySpark to develop scalable data processing solutions in Azure Databricks, integrating with Azure SQL and MongoDB to meet diverse data needs.
- Developed logical data models and data flow designs, performing quality testing to ensure accurate data conversion, and identifying root causes of issues.
- Wrote audit and logging scripts in T-SQL to track data processing activities and troubleshoot issues.
- Designed partitioned tables and indexed views in SQL Server for large fact tables, enhancing reporting performance.
- Performed T-SQL based data profiling and cleansing to standardize inconsistent source data before loading into downstream systems.
- Integrated Nielsen and IRI retail scan data into Azure Data Lake Storage, standardizing category and brand-level feeds for downstream analytics.
- Developed data ingestion and extraction pipelines using Scala and Azure Data Factory for migrating data from SQL Server to Synapse, handling SCD types 0, 1, and 2.
- Designed and deployed Azure Synapse SQL views to support self-service reporting of Nielsen and IRI metrics across regions and channels.
- Implemented Azure Databricks notebooks to handle complex file transformations that couldn't be processed in Azure Data Factory.
- Leveraged Azure Transparent Data Encryption (TDE) to secure sensitive data within the cloud and collaborated on SQL Server Reporting Services (SSRS) reports for critical insights.
- Enforced data governance and compliance best practices, including encryption and anonymization, ensuring regulatory compliance in Azure.
- Developed data ingestion and extraction pipelines using Scala and Apache Spark, leveraging distributed computing capabilities for large-scale data processing and analysis.

Avon Technologies Pvt Ltd Hyd India
Data Engineer

November 2019 to December 2021

- Analyzed source data and collaborated with business users and developers to develop the data model.
- Designing and deploying Azure Solutions using Azure data factory, synapse, Azure data bricks, Azure data flows, storage accounts
- Setup and build Azure infrastructure various resources, key vaults, azure SQL DB, IAM, synapse.
- Developed, deployed, and managed event driven and scheduled ADF pipelines to be triggered in response to events on various Azure sources including logging.
- Deployed codes using Azure Devops by merging feature branch to master, raising PR and executing CI CD pipelines
- Wrote Python scripts to load data from Web APIs to staging DB.
- Optimized Synapse queries using T-SQL query hints, result set caching, and materialized views to accelerate reporting.
- Integrated ADF parameters and variables with T-SQL procedures to support reusable, metadata-driven pipeline execution.
- Created audit and control tables with T-SQL to track pipeline execution status, error handling, and row counts within ADF.
- Collaborated with BI teams to provide optimized T-SQL views and datasets in Synapse, supporting Power BI models.
- Implemented exploratory data analysis by utilizing simple machine learning algorithm.
- Performed data analysis and profiling of source data to better understand the sources.
- Managed multiple activities in Azure data factory

- Created Azure SQL DB and managing policies for Azure SQL DB and Utilized Azure SQL DB for storage and backup on Azure. Extensive knowledge in migrating applications from internal data center to Azure.

IBing Software Solutions, Hyderabad, India

Dec 2017 to November 2019

SQL Developer

- Created database objects like tables, views, Materialized views, procedures, packages using Oracle tools like PL/SQL.
- Involved in database development by creating Functions, Procedures, Triggers, and Packages using TOAD for accessing, inserting, modifying and deleting data in the database.
- Created views and stored procedures using SQL analytical functions for data migration and transformation within SSIS packages.
- Identified and documented detailed business rules and use cases based on thorough requirements analysis.
- Participated in Performance Tuning of SQL queries using Explain Plan to improve the performance of the application.
- Worked extensively on exception handling for handling errors using system defined exceptions and user defined exceptions.
- Created ETL packages for data conversion using various transformation tasks to automate data loading and processing.
- Developed recurring automated reports using Python, visualizing data on BI platforms for streamlined analysis and decision-making.
- Contributed to JRD sessions to gather and define functional requirements, ensuring alignment with business needs.
- Implemented Agile methodology to handle frequent changes in client requirements, enabling parallel development and testing.
- Worked collaboratively within and across development and project teams in a fast-paced environment using Agile BI design principles.
- Generated periodic reports based on statistical analysis from various time frames and divisions, utilizing SSRS for clear insights.

Education details:

- Masters in Computer science, 2023 from Bradley University, Peoria, Illinois, USA
- Bachelor of Technology - 2018 from Jawaharlal Nehru Technological university, Kakinada, Andhra Pradesh, India

Certification: Fabric Data engineer Associate, DP-700